Cove-edge-induced vibrational confinement and weakened collective phonon transport in graphene nanoribbons

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Periodic cove edges alter heat conduction in graphene nanoribbons in ways that are not fully captured by geometric boundary scattering. We investigate this effect using a graphene-nanoribbon-specific neuroevolution potential together with molecular dynamics, phonon Monte Carlo simulations, lattice dynamics, and full linearized phonon Boltzmann transport calculations. Cove-edged graphene nanoribbons exhibit lower thermal conductivity than pristine armchair nanoribbons over the length and width ranges considered. The conductivity reduction obtained from molecular dynamics is larger than that predicted by the semiclassical Monte Carlo model. The residual fraction is greatest at short lengths and small widths, whereas the Monte Carlo boundary contribution becomes relatively more important as either dimension increases. Participation ratios and real-space eigenvectors identify partially confined and strongly edge-confined modes in the low- and intermediate-frequency ranges, showing how the cove geometry reconstructs the vibrational eigenstates beyond a purely geometric boundary-scattering picture. The cove-edged ribbons also show a smaller collective correction beyond the relaxation-time approximation and a lower transport-weighted Normal-to-Umklapp scattering-rate ratio. These results indicate that cove-edge patterning changes both the vibrational eigenstates and the intrinsic scattering balance, thereby reducing the mode-coupling correction beyond the independent-mode approximation.

Keywords: Graphene nanoribbons; thermal conductivity; edge-confined vibrational modes; collective phonon transport; Normal and Umklapp processes; neuroevolution potential

I. INTRODUCTION

Controlling heat conduction in low-dimensional materials requires understanding not only the magnitude of thermal conductivity but also the microscopic mechanisms through which phonons transport energy. Graphene nanoribbons (GNRs) provide a representative platform for this purpose because their thermal and electronic properties can be strongly tailored through lateral confinement and atomic edge engineering [1–4]. To reduce thermal conductance and improve thermoelectric performance, several structural strategies have been proposed for GNRs, including defect or topological-line-defect engineering, double-bend geometries, and isotope-doped GNR superlattices [5–7]. In pristine graphene, heat is predominantly carried by low-frequency acoustic phonons with long mean free paths [8–10]. Once graphene is patterned into nanoribbons, however, atomic-scale edge morphology becomes a decisive factor governing phonon propagation. Atomically sharp edges, reconstructed boundary units, edge roughness, and nanopatterned pores can introduce phonon-boundary scattering, modify the vibrational spectrum, and generate spatially confined vibrational states [11–19]. Previous studies have therefore interpreted edge-induced reductions in thermal conductivity mainly in terms of additional boundary resistance and the suppression of efficiently propagating phonon modes. Whether an atomically structured edge can also modify the intrinsic phonon collision landscape and the collective character of heat transport remains much less understood.

This question is particularly important for systems in which momentum-conserving Normal (N) processes produce an appreciable collective correction beyond the conventional independent-mode relaxation-time approximation (RTA). When N processes dominate over momentum-relaxing Umklapp (U) processes, phonons do not relax independently but can become strongly coupled through the collision operator, giving rise to collective contributions beyond the independent-mode RTA description [20–24]. Such behavior has been actively investigated in high-quality low-dimensional and layered materials [25, 26]. Atomic edge modulation may influence this regime through two related mechanisms. In addition to reflecting propagating phonons at the external boundary, an atomically structured edge can reconstruct the vibrational eigenstates, alter their group velocities and spatial character, and modify the available three-phonon scattering phase space. Edge-confined vibrations may also hybridize with extended acoustic modes and change the relative importance of N and U scattering. A direct connection among atomic edge geometry, confined vibrational modes, and the collective correction retained by the full linearized phonon Boltzmann transport equation (LBTE) has not yet been systematically established for GNRs.

Cove-edged graphene nanoribbons (Cove-GNRs) provide a controlled model system for addressing this problem. Their periodically modulated edges introduce deterministic atomic-scale variations in local width and boundary orientation while preserving translational peri-

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odicity along the nanoribbon, allowing edge-induced vibrational reconstruction to be distinguished from purely stochastic roughness scattering. Carbon-backbone motifs related to Cove-GNRs have become experimentally accessible through bottom-up solution and on-surface synthetic strategies, which enable control over ribbon width, edge periodicity, and chemical structure [27, 28]. Experimentally synthesized Cove-GNRs have exhibited reduced and tunable bandgaps, while the chirality and arrangement of cove and zigzag edge segments can significantly influence their carrier-transport properties [28, 29]. Related atomically precise donor–acceptor cove-edge nanoribbons have also been investigated as electron-accepting materials, highlighting their relevance to nanoscale electronic, optoelectronic, and photovoltaic applications [30]. From a thermal-transport perspective, previous studies have shown that edge disorder, lateral confinement, and boundary reconstruction can substantially reduce the thermal conductivity of GNRs [12–15, 17]. Nevertheless, the phonon heat transport of Cove-GNRs remains much less understood than their electronic properties, particularly regarding whether periodic cove-edge modulation merely increases geometric boundary resistance or also reshapes the competition between momentum-conserving and momentum-relaxing scattering. The carbon-only, unpassivated structures considered here are idealized models designed to isolate the influence of the cove-shaped carbon backbone; effects associated with chemical termination, substrates, and environmental interactions are beyond the scope of the present work.

Resolving this issue requires a framework capable of describing both finite-size heat conduction and mode-resolved anharmonic phonon scattering in nanoribbons with relatively large and structurally complex unit cells. First-principles lattice dynamics combined with the phonon Boltzmann transport equation provides direct access to phonon dispersions, lifetimes, scattering channels, and lattice thermal conductivity [31–33]. Its direct application to Cove-GNRs, however, is computationally demanding because the extraction of high-order interatomic force constants rapidly becomes prohibitive as the unit-cell size and edge complexity increase [34, 35]. Molecular dynamics can access substantially larger spatial and temporal scales but does not directly separate mode-specific N and U scattering or resolve the collective correction contained in the full phonon collision operator. Semiclassical phonon Monte Carlo (MC) simulations can provide a controlled description of particle-like geometric boundary scattering, but they do not explicitly capture the reconstruction of vibrational eigenstates caused by the cove geometry. Machine-learning interatomic potentials, particularly the neuroevolution potential (NEP) [36, 37], bridge these descriptions by providing near-first-principles accuracy at a computational cost compatible with large-scale molecular dynamics, force-constant extraction, lattice-dynamics analysis, and full phonon Boltzmann transport calculations.

In this work, we construct a GNR-specific NEP by fine-tuning the pretrained NEP89 potential [38] using additional structures containing armchair, zigzag, and cove-shaped edge environments. Molecular-dynamics and phonon Monte Carlo results are compared to determine how much of the cove-edge-induced conductivity reduction is reproduced by geometric boundary scattering. Participation ratios and real-space eigenvectors are used to characterize the resulting vibrational modes, while full-LBTE and RTA conductivities, together with transport-weighted N and U scattering rates, are used to assess the correction arising from mode-to-mode coupling. The calculations show that cove-edge patterning lowers the thermal conductivity, produces edge-confined vibrational modes, and reduces both the Normal-to-Umklapp scattering-rate ratio and the collective correction beyond the RTA description.

II. METHODOLOGY

A. Construction of the NEP-GNRs potential

A graphene-nanoribbon-specific neuroevolution potential (NEP), denoted as NEP-GNRs, was constructed by fine-tuning the pretrained NEP89 model [38]. The additional dataset contained monolayer graphene nanoribbons with armchair, zigzag, and cove-shaped edges, different ribbon widths, and locally perturbed atomic environments. Random atomic displacements of approximately 0.01–0.05 Å were applied to the initial structures to sample local distortions associated with different edge configurations.

All nanoribbon models considered in this work were carbon-only structures with unpassivated edges. The same edge-termination convention was used for the AGNR and Cove-GNR systems so that their differences originate from the carbon-backbone geometry rather than from different chemical terminations. The present models therefore isolate the intrinsic influence of cove-edge patterning on phonon transport; the effects of hydrogen passivation, functional groups, substrates, and environmental interactions are beyond the scope of this work. Isotope disorder was not included in any of the simulations; all carbon atoms were assigned the same atomic mass so that the comparison between AGNRs and Cove-GNRs focuses on the effect of edge geometry. All structures were fully relaxed before the molecular-dynamics, lattice-dynamics, and phonon-transport calculations.

To further enrich the dataset with finite-temperature structural perturbations, short ab initio molecular dynamics simulations were performed for two representative GNR structures at 300 K. The AIMD simulations employed a time step of 0.5 fs and contained 100 ionic steps, corresponding to a total duration of 50 fs. Twenty configurations were uniformly extracted from each trajectory, yielding 40 AIMD configurations. These short trajectories were used to introduce thermally perturbed

local environments into the fine-tuning dataset rather than to provide statistically converged finite-temperature sampling. Configurations from the datasets of Wang et al. [39] and Liang et al. [40] were also incorporated after consistency checks and removal of structures with unphysical local geometries or anomalous reference values. The final fine-tuning dataset contained 713 configurations.

Reference energies, atomic forces, and virial components were calculated using density functional theory as implemented in VASP. The Perdew–Burke–Ernzerhof exchange–correlation functional, a plane-wave cutoff energy of 600 eV, and an automatic k-point sampling scheme with `KSPACING = 0.20` Å^{-1} were employed. The electronic convergence criterion was set to 10^{-6} eV. The detailed AIMD and single-point DFT input settings are provided in Sections S1 and S2 of the Supplementary Information.

The pretrained NEP89 model was fine-tuned using GPUMD for 20,000 generations, with checkpoint potentials saved every 2,000 generations. The predicted thermal conductivity reached a stable plateau after approximately 18,000 generations, and the corresponding checkpoint potential was therefore selected for the subsequent calculations. The same NEP-GNRs model was used consistently for molecular dynamics simulations, force-constant fitting, lattice-dynamics calculations, and phonon Boltzmann transport calculations. The complete fine-tuning input and hyperparameters are provided in Section S3 of the Supplementary Information.

B. Molecular-dynamics thermal transport calculations

The length-dependent thermal conductivity was calculated at 300 K using a spectral reconstruction method combining homogeneous nonequilibrium molecular dynamics (HNEMD) with the nonequilibrium molecular dynamics (NEMD)-based spectral heat-current method. The diffusive spectral thermal conductivity, $\kappa_{\text{diff}}(\omega)$, was obtained from HNEMD simulations, whereas the spectral ballistic conductance, $G(\omega)$, was evaluated from NEMD simulations. Their ratio defines the effective frequency-dependent phonon mean free path,

$$\lambda(\omega) = \frac{\kappa_{\text{diff}}(\omega)}{G(\omega)}. \tag{1}$$

Since $\kappa_{\text{diff}}(\omega)$ and $G(\omega)$ are obtained directly from the atomistic model of each nanoribbon, the influence of the edge geometry is intrinsically incorporated into $\lambda(\omega)$.

The thermal conductivity of a nanoribbon with transport length L was then reconstructed as

$$\kappa_{\text{MD}}(L) = \int_0^\infty \frac{d\omega}{2\pi} \frac{\kappa_{\text{diff}}(\omega)}{1 + \lambda(\omega)/L}. \tag{2}$$

Throughout this work, the AGNR width W was defined as the transverse distance between the two straight opposing edges. For Cove-GNRs, the local width varies periodically along the transport direction; therefore, the effective width used for comparison with AGNRs was defined as $W = (W_{\text{min}} + W_{\text{max}})/2$, where W_{min} and W_{max} are the minimum and maximum transverse distances between the two opposing edges, respectively. The same effective thickness, $t = 0.335$ nm, was used in the MD and MC calculations, and the nominal cross-sectional area was defined as $A = Wt$ for both edge geometries.

C. Phonon Monte Carlo simulations and residual thermal suppression

Phonon MC simulations were performed at 300 K as a complementary semiclassical model for evaluating the thermal-conductivity reduction caused by geometric boundary scattering. To isolate the influence of boundary geometry, the same mode-resolved phonon properties of pristine graphene, including the phonon frequencies, group velocities, relaxation times, and mode heat capacities, were used for both AGNR and Cove-GNR simulations. Consequently, the difference between their MC thermal conductivities originates from the different boundary geometries rather than from edge-induced changes in the intrinsic phonon spectrum or phonon–phonon scattering rates.

Phonon particles were characterized by their mode index $\lambda = (\mathbf{q}, s)$, angular frequency ω_λ, group velocity $\mathbf{v}_\lambda$, relaxation time τ_λ, and mode heat capacity. Intrinsic phonon scattering was treated within the relaxation-time approximation. No additional isotope-scattering channel was included in the MC simulations. The probability that a phonon in mode λ undergoes intrinsic scattering during a time interval Δt was calculated as

$$P_\lambda^{\text{int}} = 1 - \exp\left(-\frac{\Delta t}{\tau_\lambda}\right). \tag{3}$$

Each simulation employed 2.0×10^5 computational phonon particles and was advanced for 5.0×10^4 time steps with $\Delta t = 0.005$ ns, corresponding to a total simulated duration of 250 ns.

For AGNRs, the lateral boundaries were represented by straight ribbon edges, whereas the periodically modulated cove-shaped profile was explicitly incorporated into the Cove-GNR geometry. At each boundary collision, the probability of specular reflection was determined using the Ziman–Soffer relation,

$$p_{\text{spec}} = \exp\left[-4\eta^2 k^2 \cos^2\theta\right], \tag{4}$$

where $\eta = 0.1$ nm is the boundary roughness parameter, k is the magnitude of the incident phonon wave vector, and θ is the angle between the incident wave vector and the local boundary normal. The deterministic cove-shaped boundary profile was explicitly resolved in the MC geometry. The parameter $\eta = 0.1$ nm therefore does not represent the cove modulation itself, but phenomenologically accounts for unresolved atomic-scale deviations

from the idealized boundary profile. The same roughness parameter was used for the AGNR and Cove-GNR calculations so that their difference remained primarily associated with the deterministic boundary geometry. Diffuse reflection was applied with probability $1 - p_{\mathrm{spec}}$. Following a diffuse boundary event, the outgoing phonon mode was resampled from the allowed modes propagating back into the computational domain.

Because the MC model uses the same pristine-graphene phonon properties for both edge geometries and treats phonons as semiclassical propagating particles, it captures intrinsic relaxation, finite-size transport, and geometric boundary reflection but does not explicitly include the reconstruction of vibrational eigenstates caused by cove-edge patterning. The MC results therefore provide a controlled reference for the conductivity suppression expected from particle-like boundary scattering.

The total edge-induced conductivity reduction obtained from molecular dynamics and the corresponding semiclassical boundary-scattering contribution were defined as

$$\Delta\kappa_{\mathrm{total}} = \kappa_{\mathrm{MD}}^{\mathrm{AGNR}} - \kappa_{\mathrm{MD}}^{\mathrm{Cove}}, \tag{5}$$

$$\Delta\kappa_{\mathrm{bd}} = \kappa_{\mathrm{MC}}^{\mathrm{AGNR}} - \kappa_{\mathrm{MC}}^{\mathrm{Cove}}. \tag{6}$$

The residual suppression beyond the semiclassical boundary-scattering model was then evaluated as

$$\Delta\kappa_{\mathrm{res}} = \Delta\kappa_{\mathrm{total}} - \Delta\kappa_{\mathrm{bd}}, \tag{7}$$

and the corresponding relative fractions were calculated from

$$\eta_{\mathrm{bd}} = \frac{\Delta\kappa_{\mathrm{bd}}}{\Delta\kappa_{\mathrm{total}}}, \qquad \eta_{\mathrm{res}} = \frac{\Delta\kappa_{\mathrm{res}}}{\Delta\kappa_{\mathrm{total}}}. \tag{8}$$

Here, $\Delta\kappa_{\mathrm{res}}$ denotes the part of the conductivity reduction remaining after subtracting the semiclassical geometric-boundary contribution. Since the MC model uses pristine-graphene phonon properties for both edge geometries, this residual contains edge-induced atomistic effects included in MD but absent from the MC description, including changes in vibrational eigenstates, phonon dispersion, group velocities, local force constants, anharmonic scattering rates, and phonon lifetimes. The vibrational character of the nanoribbon modes is examined below using participation-ratio and real-space eigenmode analyses.

D. Lattice dynamics and phonon Boltzmann transport calculations

Harmonic and third-order interatomic force constants were obtained using the NEP-GNRs potential through the NEP-kappa workflow [35]. Before force-constant generation, each structure was relaxed using the trained potential. A $4 \times 1 \times 1$ supercell was employed, and 500 randomly displaced structures were generated with a displacement standard deviation of 0.01 Å and a minimum interatomic distance of 1.1 Å. The cutoff radii for the second- and third-order force constants were 7.0 and 6.0 Å, respectively. The force constants were fitted using hiPhive, and the phonon transport properties were calculated on a $51 \times 1 \times 1$ wave-vector mesh over the temperature range from 100 to 1000 K in steps of 100 K. The complete NEP-kappa input parameters are provided in Section S4 of the Supplementary Information.

The lattice thermal conductivity was calculated using Phono3py under both the relaxation-time approximation and the direct solution of the full linearized phonon Boltzmann transport equation following the method of Chaput [33, 41]. Because the nanoribbon simulation cells contain vacuum in the nonperiodic directions, the RTA and full-LBTE thermal conductivities were renormalized by the factor $A_{\mathrm{cell}}/(Wt)$, where A_{cell} is the total simulation-cell cross-sectional area including vacuum, W is the AGNR width or the effective Cove-GNR width defined above, and $t = 0.335$ nm is the effective graphene thickness. Isotope scattering was not included, and the collision operator therefore contained intrinsic three-phonon scattering processes only. The RTA neglects the off-diagonal coupling of the phonon collision operator and treats each phonon mode as relaxing independently, whereas the full-LBTE solution retains the mode-to-mode coupling contained in the three-phonon collision matrix.

Recent first-principles analysis has shown that the ratio $\kappa_{\mathrm{LBTE}}/\kappa_{\mathrm{RTA}}$ provides a useful measure of collective phonon-transport effects beyond the independent-mode RTA description [42]. The difference between the full-LBTE and RTA solutions reflects the contribution of mode-to-mode coupling neglected by the RTA. Here, we quantify this relative collective correction as

$$\eta_{\mathrm{c}} = \frac{\kappa_{\mathrm{LBTE}} - \kappa_{\mathrm{RTA}}}{\kappa_{\mathrm{LBTE}}} = 1 - \frac{\kappa_{\mathrm{RTA}}}{\kappa_{\mathrm{LBTE}}}. \tag{9}$$

The quantity η_{c} is used as a measure of the relative collective correction arising from mode-to-mode coupling. A larger η_{c} corresponds to a stronger collective correction beyond the independent-mode RTA description.

Three-phonon scattering processes were decomposed into Normal and Umklapp contributions according to crystal-momentum conservation,

$$\mathbf{q}_1 \pm \mathbf{q}_2 - \mathbf{q}_3 = \mathbf{G}, \tag{10}$$

where processes with $\mathbf{G} = \mathbf{0}$ were classified as Normal processes and those with $\mathbf{G} \neq \mathbf{0}$ as Umklapp processes. To minimize representation-dependent differences in the Normal/Umklapp classification, the AGNR and Cove-GNR structures were represented using the same axial periodicity and reciprocal-lattice convention when their scattering processes were compared.

To emphasize phonon modes with larger heat-capacity and group-velocity weights, the Normal and Umklapp

scattering rates were averaged using the factor $C_\lambda|\mathbf{v}_\lambda|^2$. The transport-weighted average scattering rate was defined as

$$\langle\Gamma_X\rangle = \frac{\sum_\lambda C_\lambda \left|\mathbf{v}_\lambda\right|^2 \Gamma_{X,\lambda}}{\sum_\lambda C_\lambda \left|\mathbf{v}_\lambda\right|^2}, \qquad X = N, U, \tag{11}$$

where C_λ and $\mathbf{v}_\lambda$ are the heat capacity and group velocity of phonon mode λ, respectively, and $\Gamma_{N,\lambda}$ and $\Gamma_{U,\lambda}$ are its Normal and Umklapp three-phonon scattering rates. The corresponding transport-weighted Normal-to-Umklapp ratio was defined as $\langle\Gamma_N\rangle/\langle\Gamma_U\rangle$. This ratio characterizes the relative predominance of momentum-conserving and momentum-relaxing scattering among phonon modes with larger heat-capacity and group-velocity weights. Because this ratio alone does not uniquely determine collective transport, it was interpreted together with the difference between the full-LBTE and RTA thermal conductivities.

The spatial character of each phonon mode was quantified using the participation ratio,

$$P(\mathbf{q}, s) = \frac{\left[\sum_i \sum_\alpha \left|e_{i\alpha}(\mathbf{q}, s)\right|^2\right]^2}{N \sum_i \left(\sum_\alpha \left|e_{i\alpha}(\mathbf{q}, s)\right|^2\right)^2}, \tag{12}$$

where N is the number of atoms in the unit cell and $e_{i\alpha}(\mathbf{q}, s)$ is the normalized eigenvector component of atom i along Cartesian direction α. Values close to unity correspond to modes extended throughout the unit cell, whereas smaller values indicate vibrations confined to a smaller subset of atoms. The participation ratios were analyzed together with real-space eigenvectors to distinguish extended, partially confined, and strongly edge-confined vibrational modes.

Compared with conventional DFT-based finite-displacement calculations, the NEP-kappa workflow avoids repeated DFT force evaluations during force-constant extraction; previous benchmarks showed DFT-level phonon accuracy with a computational-cost reduction of several orders of magnitude [34, 35].

III. RESULTS AND DISCUSSION

A. Validation of the NEP-GNRs potential

Because the same NEP-GNRs potential was used for molecular dynamics simulations, force-constant fitting, lattice-dynamics analysis, and phonon Boltzmann transport calculations, its reliability was assessed at several complementary levels. We examined the structural coverage of the fine-tuning dataset, the accuracy of the predicted energetics, the harmonic phonon properties of graphene and edge-containing nanoribbons, and the stability of the calculated thermal conductivity during fine-tuning.

Figure 1(a) shows a principal component analysis (PCA) projection of the atomic descriptors included in the fine-tuning dataset. The configurations span a broad region of descriptor space and cover local environments associated with pristine graphene-like structures, armchair and zigzag edges, cove-shaped boundaries, and thermally or randomly perturbed configurations. This structural diversity is important because the local coordination and bonding environments near reconstructed edges differ appreciably from those in bulk-like graphene. The PCA distribution therefore indicates that the fine-tuning dataset covers both interior and edge atomic environments encountered in the subsequent AGNR and Cove-GNR calculations.

The convergence of the fine-tuning process was assessed through the calculated thermal conductivity of pristine graphene, as shown in Fig. 1(b). Starting from the pretrained NEP89 model, the predicted thermal conductivity changes rapidly during the early stages of fine-tuning and subsequently reaches a stable plateau. Beyond approximately 18,000 generations, further training produces no appreciable variation in the calculated conductivity. The value obtained from the converged model also falls within the range of representative first-principles and experimental results reported for graphene, although the literature values span a broad interval because of differences in sample size, isotope composition, defect concentration, and measurement or simulation conditions. The checkpoint obtained after 18,000 generations was therefore selected for the subsequent calculations.

Figure 1(c) evaluates the energetic accuracy of NEP-GNRs for the configurations included in the parity analysis. The predicted energies closely follow the DFT reference values over the full energy range, with an MAE of approximately 27.0 meV atom^{-1} and an RMSE of approximately 34.0 meV atom^{-1}. The absence of pronounced systematic deviations indicates that the potential consistently describes configurations containing both graphene-like interior environments and reconstructed edge structures.

The harmonic accuracy of the potential is evaluated in Figs. 1(d) and 1(e). For pristine graphene, the NEP-GNRs phonon dispersion reproduces the overall acoustic and optical branch structure and follows the reference DFT frequencies along the Γ–M–K–Γ path. The agreement in the low-frequency acoustic branches is important for thermal-transport calculations because these modes make a substantial contribution to heat conduction. For the representative AGNR, the NEP-GNRs dispersion also follows the DFT reference along the Γ–X path, indicating that the potential describes the harmonic vibrational properties of nanoribbons with free edges. Together with the PCA coverage of cove-edge environments and the DFT-level energetic validation, these phonon comparisons support the use of NEP-GNRs for the structural, vibrational, and thermal-transport calculations performed below.

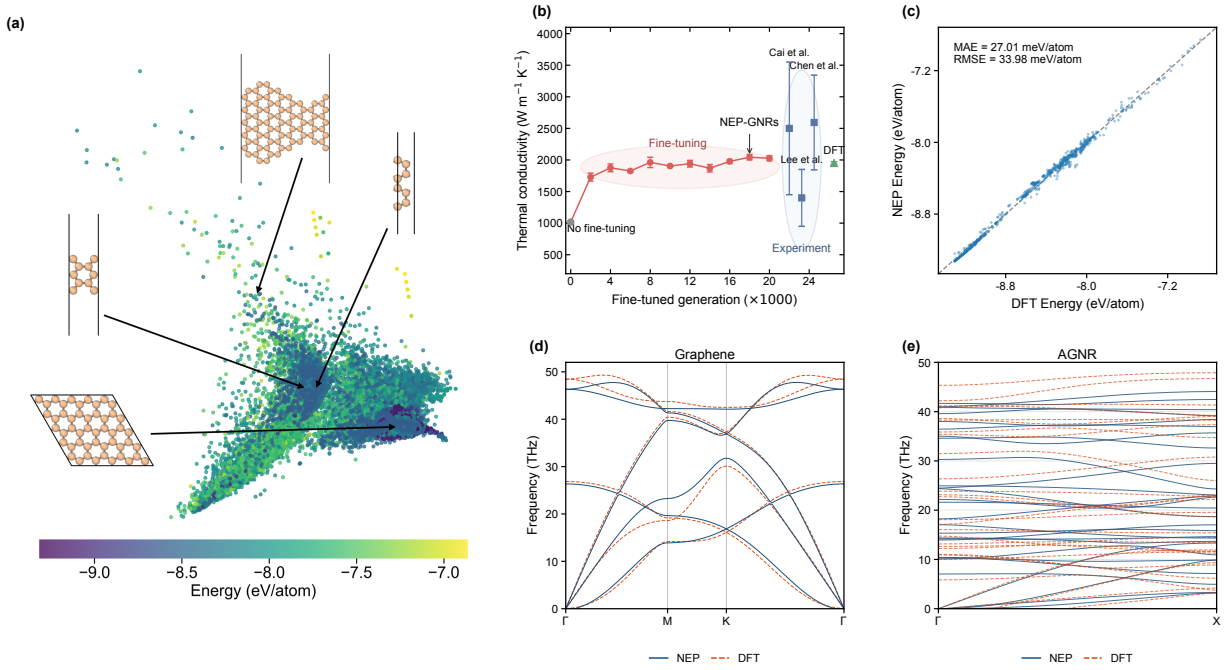


FIG. 1. Validation of the fine-tuned NEP-GNRs potential for graphene and graphene-nanoribbon edge environments. (a) Principal component analysis projection of the atomic descriptors in the fine-tuning dataset. (b) Evolution of the predicted thermal conductivity of pristine graphene at 300 K during fine-tuning from the pretrained NEP89 model to NEP-GNRs, together with representative DFT and experimental values reported in the literature [43–45]. (c) Parity plot comparing the energies predicted by NEP-GNRs with the corresponding DFT reference values. MAE and RMSE denote the mean absolute error and root-mean-square error, respectively. (d) Phonon dispersion of pristine graphene predicted by NEP-GNRs compared with the reference DFT result along the Γ–M–K–Γ path. (e) Phonon dispersion of a representative armchair graphene nanoribbon predicted by NEP-GNRs compared with the DFT reference result along the Γ–X path.

B. Size- and temperature-dependent thermal conductivity and residual edge-induced suppression

The atomic structures considered in this work are shown in Figs. 2(a) and 2(b). A pristine AGNR with straight, periodic, and unpassivated armchair edges serves as the reference structure, whereas the Cove-GNR contains periodically modulated unpassivated cove-shaped boundaries. Here, "pristine AGNR" refers to the ideal straight-edged nanoribbon without cove modulation, additional edge disorder, or chemical functionalization. The same edge-termination convention is used for both systems, allowing the effect of the carbon-backbone geometry to be isolated from that of chemical passivation. Compared with the straight armchair edge, the cove geometry introduces deterministic atomic-scale variations in the local width and boundary orientation. For the Cove-GNR, the geometric modulation is characterized by the cove period P_{cove}, the cove depth d_{cove}, and the local vertex angle θ, as indicated in Fig. 2(b). The minimum and maximum transverse distances between the two opposing edges are denoted by W_{min} and W_{max}, respectively, and the effective width used for comparison with the AGNR is defined as $W = (W_{\mathrm{min}} + W_{\mathrm{max}})/2$.

Figure 2(c) compares the length-dependent thermal conductivity obtained from the MD-based spectral reconstruction and phonon MC simulations at a fixed width of $W = 20$ nm. In all calculated cases, the thermal conductivity increases with transport length as phonons with progressively longer mean free paths contribute to heat conduction. Cove-GNRs exhibit lower thermal conductivity than pristine AGNRs over the entire length range considered, demonstrating that the periodically modulated edge strongly suppresses heat conduction. The difference between the two structures also becomes more pronounced with increasing length. Whereas the conductivity of the AGNR continues to increase substantially, the corresponding growth in the Cove-GNR is weaker. This weaker length dependence is consistent with stronger suppression of transport channels associated with relatively long phonon mean free paths.

The comparison between MD and MC helps separate the semiclassical geometric-boundary contribution from the remaining atomistic contribution. In the MC calculations, both AGNRs and Cove-GNRs were assigned the same mode-resolved phonon properties of pristine graphene, so their conductivity difference arises exclusively from the different geometric boundaries and the associated semiclassical phonon reflection. The MD simulations, in contrast, explicitly describe the atomistic vibrational dynamics of each nanoribbon and therefore include edge-induced changes in the vibrational spectrum,

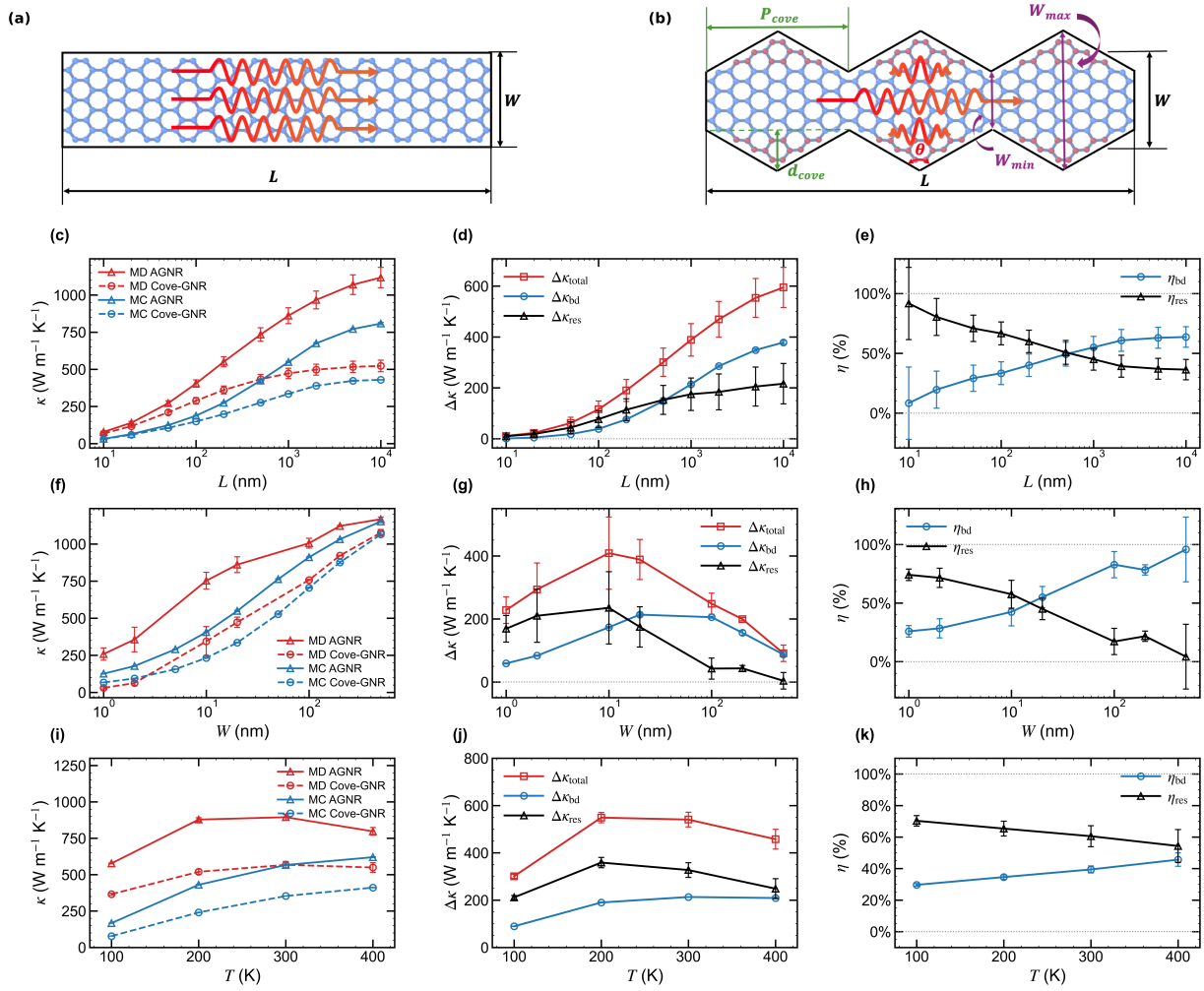


FIG. 2. Atomic structures and size- and temperature-dependent thermal-transport suppression in graphene nanoribbons with different edge morphologies. (a) Pristine AGNR with unpassivated straight armchair edges. (b) Cove-GNR with unpassivated periodically modulated edges. Here, P_{cove} denotes the cove period, d_{cove} denotes the cove depth, W_{min} and W_{max} denote the minimum and maximum transverse distances between the two opposing edges, respectively, and θ denotes the local vertex angle associated with the cove modulation. The effective ribbon width is defined as $W = (W_{\mathrm{min}} + W_{\mathrm{max}})/2$, and the AGNR and Cove-GNR are compared at the same W. (c) Length-dependent thermal conductivity κ at a fixed width of $W = 20$ nm obtained from MD-based spectral reconstruction and phonon MC simulations. (d) Length-dependent comparison of the total MD reduction $\Delta\kappa_{\mathrm{total}}$, the boundary reduction predicted by the semiclassical MC model $\Delta\kappa_{\mathrm{bd}}$, and the resulting effective atomistic residual $\Delta\kappa_{\mathrm{res}}$. (e) Corresponding normalized model terms η_{bd} and η_{res} as functions of length. (f) Width-dependent thermal conductivity at a fixed length of $L = 1000$ nm. (g) Corresponding width-dependent comparison of $\Delta\kappa_{\mathrm{total}}$, $\Delta\kappa_{\mathrm{bd}}$, and $\Delta\kappa_{\mathrm{res}}$. (h) Normalized model terms η_{bd} and η_{res} as functions of width. (i) Temperature-dependent thermal conductivity at fixed ribbon dimensions of $W = 50$ nm and $L = 1000$ nm. (j) Temperature-dependent comparison of $\Delta\kappa_{\mathrm{total}}$, $\Delta\kappa_{\mathrm{bd}}$, and the effective atomistic residual $\Delta\kappa_{\mathrm{res}}$. (k) Normalized model terms η_{bd} and η_{res} as functions of temperature. In panels (c), (f), and (i), red and blue curves denote the MD-based spectral reconstruction and MC results, respectively; solid triangle-marked lines correspond to pristine AGNRs, whereas dashed circle-marked lines correspond to Cove-GNRs.

anharmonic interactions, and mode hybridization. The larger conductivity reduction obtained from MD shows that geometric boundary reflection alone is insufficient to reproduce the full influence of the cove edge.

The model-based comparison in Fig. 2(d) quantifies this difference. The total conductivity reduction, $\Delta\kappa_{\mathrm{total}}$, increases with ribbon length, indicating that the effect of the cove geometry becomes more pronounced as longer-mean-free-path phonons contribute to transport. The semiclassical boundary-scattering contribution, $\Delta\kappa_{\mathrm{bd}}$, also increases with length and becomes the largest absolute component in sufficiently long nanoribbons. By comparison, the residual suppression, $\Delta\kappa_{\mathrm{res}}$, increases more moderately and tends toward saturation.

Here, $\Delta\kappa_{\mathrm{res}}$ is defined by the difference between the MD and MC reductions. It collects the edge-induced atomistic contributions included in the MD description but absent from the semiclassical MC model,

such as changes in vibrational eigenstates, phonon dispersion, group velocities, local force constants, anharmonic scattering rates, and phonon lifetimes. The participation-ratio and eigenmode analyses below focus on the vibrational-eigenstate component of this residual response.

The relative fractions in Fig. 2(e) further illustrate the length-dependent crossover. At short lengths, the residual fraction η_{res} constitutes the larger part of the total conductivity reduction, whereas the boundary-scattering fraction η_{bd} is relatively small. As the ribbon length increases, η_{bd} rises and η_{res} decreases. This trend shows that the relative importance of suppression not captured by the semiclassical boundary model is greatest in short nanoribbons, while the semiclassical boundary term accounts for an increasing fraction of the total conductivity reduction as the transport length grows.

Figures 2(f)–2(h) show the corresponding width-dependent results at a fixed length of $L = 1000$ nm. The thermal conductivity increases with ribbon width for both AGNRs and Cove-GNRs as the boundary-to-interior ratio decreases and the relative influence of the edges becomes weaker. Nevertheless, the conductivity of the Cove-GNR remains lower than that of the AGNR throughout the studied width range, indicating that the influence of the periodically modulated edge is not restricted to the narrowest structures.

Figures 2(g) and 2(h) quantify the width dependence of the conductivity reduction and its normalized components. The total reduction, $\Delta\kappa_{\mathrm{total}}$, decreases as the ribbon becomes wider. The residual suppression, $\Delta\kappa_{\mathrm{res}}$, is most pronounced in the narrowest ribbons and decreases with increasing width, whereas the semiclassical boundary contribution, $\Delta\kappa_{\mathrm{bd}}$, remains appreciable over a broader width range.

The normalized terms reveal the corresponding crossover. In narrow nanoribbons, η_{res} exceeds η_{bd}, indicating that most of the conductivity reduction is not reproduced by the semiclassical geometric-boundary model. With increasing width, η_{res} decreases, while η_{bd} increases and eventually becomes dominant. Together with the length-dependent results, the MD–MC comparison shows that the relative model residual is largest at short lengths and small widths, whereas the geometric boundary term accounts for an increasing fraction of the total reduction as either dimension increases.

Figures 2(i)–2(k) further examine how this decomposition varies with temperature at fixed ribbon dimensions of $W = 50$ nm and $L = 1000$ nm. The MD-based spectral reconstruction and MC simulations both show a nonmonotonic temperature dependence of the thermal conductivity over 100–400 K, reflecting the competition between heat-capacity effects, phonon lifetime changes, and boundary-limited transport. The decomposition in Fig. 2(j) shows that both $\Delta\kappa_{\mathrm{bd}}$ and $\Delta\kappa_{\mathrm{res}}$ remain appreciable over this temperature range. The normalized terms in Fig. 2(k) indicate that η_{bd} increases from about 30% at 100 K to about 46% at 400 K, whereas η_{res} decreases from about 70% to about 54%. Thus, the residual edge-induced suppression is more prominent at lower temperature, while the semiclassical geometric-boundary contribution becomes increasingly important as temperature rises. This trend is consistent with enhanced intrinsic phonon–phonon scattering at higher temperature, which reduces the relative weight of long-lived phonon modes that are more sensitive to residual edge-induced spectral and dynamical modifications.

C. Edge-confined vibrational modes in Cove-GNRs

To examine vibrational features that are absent from the semiclassical boundary model, we analyze the phonon dispersions, participation ratios (PRs), and representative real-space eigenvectors of the AGNR and Cove-GNR. The PR measures the fraction of atoms involved in a phonon mode: values close to unity correspond to vibrations distributed throughout the unit cell, whereas smaller values indicate that the atomic displacements are concentrated on a smaller subset of atoms. Because the modes retain a well-defined wave vector along the periodic direction, a reduced participation ratio is interpreted as spatial confinement within the nanoribbon unit cell rather than complete localization along the transport direction.

Figure 3(a) compares the frequency-resolved average participation ratios of the two nanoribbons. The pristine AGNR maintains relatively large participation ratios over a broad frequency range, including the low-frequency region containing acoustic branches. In contrast, the Cove-GNR exhibits lower participation ratios within several low- and intermediate-frequency intervals, showing that the periodic edge modulation changes the spatial distribution of its vibrational eigenstates.

The mode-resolved dispersions in Figs. 3(b) and 3(c) provide a more detailed comparison. Many low-frequency branches in the AGNR retain large participation ratios, indicating displacement fields that extend across most of the ribbon. The Cove-GNR contains more branches with reduced participation ratios, particularly in the low- and intermediate-frequency regions. These branches indicate stronger spatial confinement, while their specific edge-centered character is examined below using representative real-space eigenvectors.

The enlarged low-frequency region in Fig. 3(d) shows that the reduction in participation ratio extends to acoustic and low-lying branches. A reduced participation ratio does not by itself determine the longitudinal group velocity or modal thermal conductivity. It nevertheless shows that the vibrational amplitude is redistributed toward a smaller part of the nanoribbon, representing a mode-level effect that is absent from the semiclassical boundary model.

The representative eigenvectors in Figs. 3(e)–3(g) illustrate three different spatial distributions. The mode in Fig. 3(e), with PR = 0.996316, has an almost uniform

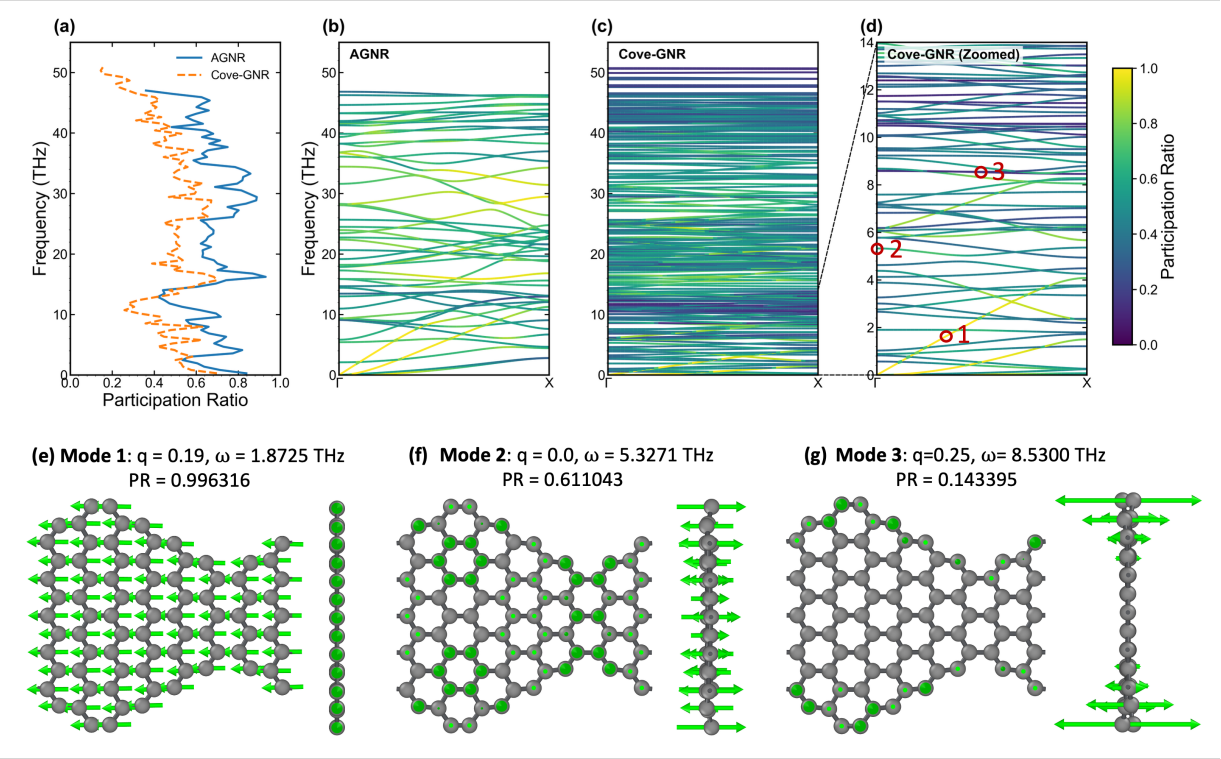


FIG. 3. Participation-ratio analysis and representative phonon eigenmodes of the AGNR and Cove-GNR. (a) Frequency-resolved average participation ratios of phonon modes in the two nanoribbons. Smaller values indicate stronger spatial confinement of the atomic displacements within the unit cell. (b) Phonon dispersion of the AGNR colored by the mode participation ratio. (c) Phonon dispersion of the Cove-GNR colored by the mode participation ratio. (d) Enlarged low-frequency region of the Cove-GNR dispersion. Red circles identify three representative modes selected for real-space eigenvector visualization. (e) Representative extended mode in the Cove-GNR at the reduced wave-vector coordinate $q = 0.19$, with $\omega/2\pi = 1.8725$ THz and PR $= 0.996316$. (f) Representative partially confined mode at the Γ point, with $q = 0$, $\omega/2\pi = 5.3271$ THz and PR $= 0.611043$. (g) Representative strongly edge-confined mode at $q = 0.25$, with $\omega/2\pi = 8.5300$ THz and PR $= 0.143395$. Green arrows represent the atomic displacement vectors. The color scale in panels (b)–(d) denotes the participation ratio, with larger values corresponding to spatially extended modes and smaller values corresponding to increasingly confined modes.

displacement field across the nanoribbon and is therefore identified as an extended mode.

The mode in Fig. 3(f) has an intermediate participation ratio of PR $= 0.611043$. Its displacement field spans a substantial part of the ribbon but is more unevenly distributed, with enhanced motion in selected edge regions. We therefore describe this mode as partially confined rather than fully localized.

For the mode in Fig. 3(g), the participation ratio decreases to PR $= 0.143395$. Most of its vibrational amplitude is concentrated on atoms near the cove-shaped boundaries, while the ribbon interior participates only weakly. This displacement pattern identifies the mode as strongly edge-confined. The real-space eigenvector is essential for this classification because a small participation ratio alone does not specify where the vibrational amplitude is located.

The participation ratios and real-space eigenvectors show that the cove geometry produces modes with different degrees of spatial confinement, including strongly edge-confined modes in the low- and intermediate-frequency ranges. The cove boundary therefore changes the vibrational eigenstates rather than acting solely as a geometric reflecting surface. This mode-level reconstruction is one way in which the cove geometry enters the atomistic response represented by $\Delta\kappa_{\mathrm{res}}$.

D. Reduced collective correction in Cove-GNRs

We next examine whether cove-edge patterning changes the correction beyond the independent-mode RTA description. The second- and third-order interatomic force constants were obtained through the NEP-kappa workflow [35] using the same NEP-GNRs potential as in the molecular-dynamics calculations. The lattice-dynamics, scattering-rate, and Boltzmann transport calculations are therefore based on a common description of the interatomic interactions.

Figures 4(a) and 4(b) compare the temperature-dependent RTA and full-LBTE thermal conductivities of the AGNRs and Cove-GNRs. For the AGNRs, the full-

LBTE conductivity is consistently higher than the RTA result, particularly at lower temperatures. Because the RTA neglects the off-diagonal coupling of the collision operator, the difference between the two solutions reflects the correction arising from mode-to-mode coupling.

For the Cove-GNRs, the difference between the full-LBTE and RTA conductivities is substantially smaller than that in the corresponding AGNRs for both $W = 1$ and 2 nm. Figure 4(c) further shows the width dependence of the relative collective correction η_c. For both edge geometries, η_c decreases as the ribbon width increases from 1 to 2 nm. At 100 K, for example, η_c decreases from approximately 0.70 to 0.53 for the AGNR and from 0.19 to 0.13 for the Cove-GNR. At 1000 K, the corresponding values decrease from approximately 0.28 to 0.18 and from 0.16 to 0.08, respectively. Importantly, the Cove-GNR retains a smaller η_c than the corresponding AGNR at every matched width and temperature, indicating that cove-edge patterning consistently weakens the collective mode-to-mode coupling over the width range considered.

To examine the scattering processes associated with this trend, the intrinsic three-phonon rates were separated into Normal and Umklapp contributions. Normal processes conserve crystal momentum within the first Brillouin zone, whereas Umklapp processes involve a nonzero reciprocal-lattice vector and relax crystal momentum. The AGNR and Cove-GNR were represented using the same axial periodicity and reciprocal-lattice convention to reduce representation-dependent differences in the Normal/Umklapp classification. In the frequency-dependent scattering-rate plots, each point corresponds to an individual phonon mode $\lambda = (\mathbf{q}, s)$. The average scattering rates were calculated using the transport-weighted mode average defined in Eq. (11).

Figures 4(d) and 4(e) show the mode-resolved Normal and Umklapp scattering rates of the 1-nm-wide AGNR at 100 and 300 K. At 100 K, the Normal rates exceed the Umklapp rates over a broad part of the low-frequency spectrum. This region contains acoustic modes that contribute substantially to heat conduction, consistent with the relatively large collective correction η_c of the AGNR. At 300 K, both rates increase, while the difference between the Normal and Umklapp contributions becomes smaller.

The transport-weighted averages in Fig. 4(f) summarize this temperature dependence. At low temperatures, $\langle\Gamma_N\rangle$ is substantially larger than $\langle\Gamma_U\rangle$. The ratio $\langle\Gamma_N\rangle/\langle\Gamma_U\rangle$ decreases with increasing temperature as the relative contribution of Umklapp scattering grows, in parallel with the reduction in η_c.

The corresponding Cove-GNR results are shown in Figs. 4(g) and 4(h). Compared with the AGNR, the separation between the Normal and Umklapp rates is smaller over much of the low- and intermediate-frequency spectrum. The altered vibrational spectrum is therefore accompanied by a change in the intrinsic three-phonon scattering landscape, with a greater relative contribution from momentum-relaxing processes.

Figure 4(i) shows that the transport-weighted $\langle\Gamma_N\rangle/\langle\Gamma_U\rangle$ ratio of the 1-nm-wide Cove-GNR remains below that of the corresponding AGNR. Normal scattering is still important, but its predominance over Umklapp scattering is reduced. This trend is consistent with the smaller collective correction η_c of the Cove-GNR.

The two indicators point in the same direction: Cove-GNRs exhibit both a smaller collective correction η_c and a lower transport-weighted $\langle\Gamma_N\rangle/\langle\Gamma_U\rangle$ ratio. Together with the participation-ratio results, these trends suggest that cove-edge-induced changes in the vibrational eigenstates and scattering phase space reduce the relative importance of collective mode-to-mode coupling. The cove modulation therefore affects both boundary scattering and the collision dynamics beyond the independent-mode RTA description.

The LBTE calculations above focus on three-phonon scattering processes. Higher-order four-phonon scattering can also contribute to the thermal resistance of strongly anharmonic or low-thermal-conductivity materials [46, 47], particularly at elevated temperatures. Quantifying this contribution in Cove-GNRs would require fourth-order force constants and remains an interesting direction for future work.

IV. CONCLUSION

In this work, we examined phonon heat transport in cove-edged graphene nanoribbons using molecular dynamics, phonon Monte Carlo simulations, lattice dynamics, and full-LBTE calculations based on the same NEP-GNRs potential. Cove-GNRs exhibit lower thermal conductivity than pristine AGNRs across the length, width, and temperature ranges considered. The MD–MC comparison shows that geometric boundary reflection captures only part of this reduction. The remaining atomistic contribution is largest at short lengths and small widths, whereas the boundary-scattering contribution becomes relatively more important as either dimension increases.

Participation-ratio and real-space eigenvector analyses show that the cove geometry changes the vibrational eigenstates and introduces partially confined and strongly edge-confined modes in the low- and intermediate-frequency ranges. The full-LBTE calculations further show that Cove-GNRs have a smaller collective correction beyond the independent-mode RTA description and a lower transport-weighted Normal-to-Umklapp scattering-rate ratio than the corresponding AGNRs.

These results indicate that cove-edge modulation affects thermal transport through both geometric boundary reflection and atomistic changes in vibrational and scattering properties. The MD–MC residual reflects this remaining atomistic contribution beyond the semiclassical boundary model, with edge-confined vi-

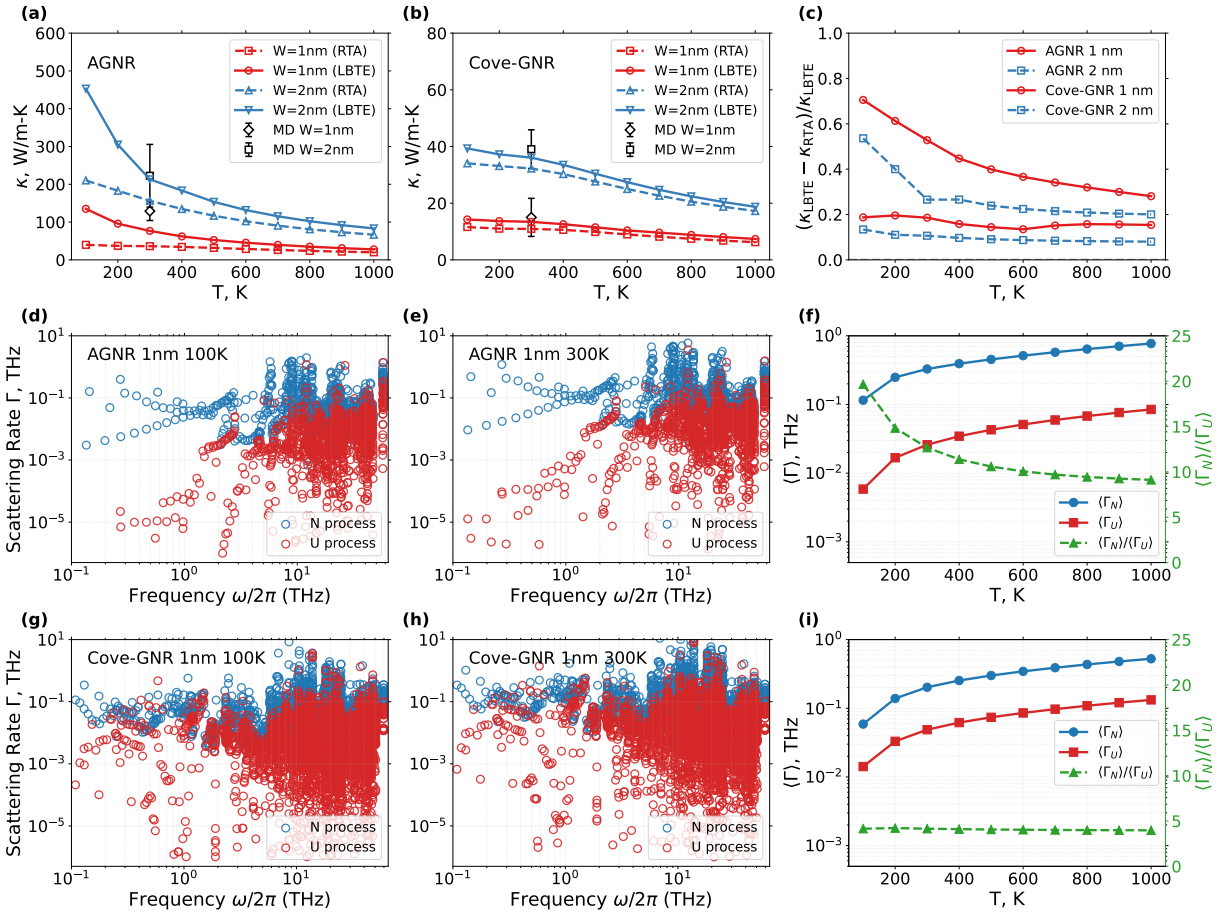


FIG. 4. Lattice thermal transport and Normal/Umklapp scattering in graphene nanoribbons with different edge geometries. (a,b) Temperature-dependent thermal conductivity κ of AGNRs and Cove-GNRs with widths of 1 and 2 nm, calculated using the relaxation-time approximation and the direct solution of the full linearized phonon Boltzmann transport equation. The black symbols in panels (a) and (b) denote MD-based thermal-conductivity values at 300 K for $W = 1$ and 2 nm, included for comparison with the LBTE/RTA results at overlapping ribbon widths. (c) Relative collective correction, $\eta_c = (\kappa_{\mathrm{LBTE}}-\kappa_{\mathrm{RTA}})/\kappa_{\mathrm{LBTE}}$, quantifying the mode-to-mode coupling correction beyond the independent-mode RTA description. (d,e) Mode-resolved frequency-dependent Normal (N) and Umklapp (U) three-phonon scattering rates in the 1-nm-wide AGNR at 100 and 300 K, respectively. (f) Transport-weighted average scattering rates $\langle\Gamma_N\rangle$ and $\langle\Gamma_U\rangle$, together with the ratio $\langle\Gamma_N\rangle/\langle\Gamma_U\rangle$, for the 1-nm-wide AGNR. (g,h) Corresponding mode-resolved frequency-dependent N and U scattering rates in the 1-nm-wide Cove-GNR at 100 and 300 K, respectively. (i) Transport-weighted average scattering rates and their ratio for the 1-nm-wide Cove-GNR.

brational modes forming one part of the response. Other effects, including phonon-dispersion changes, local force-constant modifications, anharmonic scattering, and phonon-lifetime variations, may also contribute.

Mechanical strain provides another possible route for tuning phonon transport in cove-edged GNRs by modifying phonon dispersion, group velocities, and the phase space of Normal and Umklapp scattering processes [48, 49]. Strain-dependent changes in the Normal-to-Umklapp scattering balance therefore remain an interesting direction for future work.

Appendix: Supplementary Information

The Supplementary Information provides additional computational details and supporting analyses for the results discussed in the main text. Sections S1–S4 summarize the DFT input settings, NEP fine-tuning parameters, and NEP-kappa workflow used for force-constant fitting and phonon Boltzmann transport calculations. Section S5 examines the sensitivity of the phonon Monte Carlo baseline to the boundary roughness parameter. Section S6 discusses the influence of hydrogen edge passivation on phonon dispersion and thermal conductivity. Section S7 presents the statistical convergence of the HNEMD thermal conductivity. Sections S8 and S9 provide additional mode-resolved vibrational-weight, edge-enrichment, and low-participation-mode analyses supporting the interpretation of edge-confined vibrational modes. Section S10 reports the effective phonon mean free path.

Data availability

The processed data supporting the findings of this study are available from the corresponding authors upon reasonable request.

CRediT authorship contribution statement

Fei Yin: Validation, Formal analysis, Investigation, Writing - original draft. **Gang Wang:** Validation, Formal analysis, Investigation, Writing - original draft. **Alexander A. Barinov:** Formal analysis, Writing - review & editing. **Shixian Liu:** Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing - review & editing. **Ke Xu:** Conceptualization, Methodology, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary Information

S1. The INCAR input file for AIMD simulations

To sample thermally perturbed graphene-nanoribbon configurations, ab initio molecular dynamics (AIMD) simulations were performed for two representative GNR structures using VASP at 300 K. Non-spin-polarized calculations were adopted. The ionic time step was set to 0.5 fs, and 100 ionic steps were performed for each structure, corresponding to a total simulation time of 50 fs per trajectory. The following INCAR settings were used for the GNR AIMD calculations.

```
Graphene AIMD
ISTART = 1
ISPIN = 1
ICHARG = 2           # Start from the initial charge density
LREAL = Auto         # Projection operators: automatic
ENCUT = 600          # Cut-off energy for plane-wave basis set, in eV
PREC = Accurate      # Precision level: high
LWAVE = .FALSE.      # Do not write WAVECAR
LCHARG = .FALSE.     # Do not write CHGCAR
ADDGRID = .TRUE.     # Use an additional support grid
LASPH = .TRUE.       # Include non-spherical contributions inside PAW
   spheres

# Electronic relaxation
ISMEAR = -5          # Tetrahedron smearing method with Blochl corrections
SIGMA = 0.1          # Width of smearing, in eV
EDIFF = 1E-06        # SCF energy convergence criterion, in eV
ALGO = Fast          # Electronic minimization algorithm
NELM = 60            # Maximum number of electronic SCF steps

# Molecular dynamics
IBRION = 0           # Activate molecular dynamics
NSW = 100            # Number of ionic steps
EDIFFG = -1E-02
POTIM = 0.5          # Time step, in fs
TEBEG = 300          # Initial temperature, in K
TEEND = 300          # Final temperature, in K
SMASS = -3
MDALGO = 3           # Langevin dynamics algorithm
ISYM = 0             # Disable symmetry
ISIF = 2             # Calculate stress tensor while keeping the cell fixed
```

S2. The INCAR input file for single-point reference calculations

Single-point DFT calculations were carried out for selected configurations to generate reference energies, atomic forces, and virial information for NEP fine-tuning. During these calculations, the ionic positions were fixed and only the electronic degrees of freedom were optimized. A higher-accuracy setting was used compared with AIMD sampling. The following INCAR settings were used for the single-point reference calculations.

```
GGA = PE             # Use the PBE functional for exchange-correlation
LREAL = Auto          # Projection operators: automatic
ENCUT = 600           # Cut-off energy for plane-wave basis set, in eV
KSPACING = 0.2        # Automatically generate k-points
KGAMMA = .TRUE.       # Include the Gamma point
NSW = 1               # Single-step calculation
IBRION = -1           # Ions are not moved
ALGO = Normal         # Electronic minimization algorithm
EDIFF = 1E-06         # SCF energy convergence criterion, in eV
ISMEAR = -5           # Tetrahedron smearing method with Blochl corrections
SIGMA = 0.10          # Width of smearing, in eV
PREC = Accurate       # Precision level: high
NELM = 150            # Maximum number of electronic SCF steps
```

S3. Hyperparameters for NEP89 fine-tuning

The NEP model was fine-tuned from the pretrained NEP89 potential. The pretrained potential file `nep89_20250409.txt` and the corresponding restart file `nep89_20250409.restart` were used as the initial model. Because the present system contains only carbon atoms, the element type was set to C. The fine-tuning input file reads as follows.

```
# for prediction
#type       89 H He Li Be B C N O F Ne Na Mg Al Si P S Cl Ar K Ca Sc Ti V
   Cr Mn Fe Co Ni Cu Zn Ga Ge As Se Br Kr Rb Sr Y Zr Nb Mo Tc Ru Rh Pd Ag
   Cd In Sn Sb Te I Xe Cs Ba La Ce Pr Nd Pm Sm Eu Gd Tb Dy Ho Er Tm Yb Lu
   Hf Ta W Re Os Ir Pt Au Hg Tl Pb Bi Ac Th Pa U Np Pu
#prediction 1

# Fine-tuning
fine_tune nep89_20250409.txt nep89_20250409.restart
type 1 C

# # These cannot be changed:
version 4
zbl 2
cutoff 6 5               # Radial and angular cutoffs, in Angstrom
n_max 4 4                # Size of radial and angular basis
basis_size 8 8           # Number of radial and angular basis functions
l_max 4 2 1              # Expansion orders for angular terms
neuron 80                # Number of neurons in the hidden layer

# These can be changed:
lambda_1 0               # L1 regularization weight
lambda_e 1               # Weight of energy loss term
lambda_f 1               # Weight of force loss term
lambda_v 1               # Weight of virial loss term
batch 1000               # Batch size for fine-tuning
population 50            # Population size used in the SNES algorithm
save_potential 2000 1    # Save checkpoint potential every 2000 generations
generation 20000         # Total number of fine-tuning generations
```

The pretrained NEP89 model was fine-tuned for 20,000 generations. The checkpoint potential saved at 18,000 generations was selected for the subsequent molecular-dynamics simulations, force-constant calculations, lattice-dynamics analyses, and phonon Boltzmann transport calculations.

S4. The input file for NEP-kappa

The main settings used for the NEP-kappa calculations are summarized in the following input file. This file defines the input structure, NEP model, supercell size, force-constant fitting parameters, q-point mesh, temperature range, and transport method.

```
# --- Basic Settings ---
--poscar      Structure/POSCAR
--nep_model   NEP/nep_18w.txt
--do_relax    true          # Relax the structure before calculation

# --- Supercell Settings ---
--dim         4 1 1         # Supercell size along x, y, and z

# --- HiPhive Settings ---
--use_hiphive true          # Use HiPhive for force-constant fitting
--n_structures 500          # Number of rattled structures
--rattle_std   0.01         # Random displacement amplitude in Angstrom
--min_dist     1.1          # Minimum interatomic distance in Angstrom
--cutoffs      7.0 6.0      # Cutoffs for 2nd- and 3rd-order force
   constants

# --- Calculation Settings ---
--mesh        51 1 1        # q-point mesh
--temps       100 1000 100  # Temperature range: Tmin Tmax Tstep

# --- Method Selection ---
--fc2fc3      true          # Calculate 2nd- and 3rd-order force constants
--method      lbte          # Transport method: lbte or rta
--wigner      false         # false = BTE; true = WTE
```

Here, `--poscar` and `--nep_model` specify the atomic structure and the trained NEP potential, respectively. The option `--do_relax true` indicates that the structure is relaxed before force-constant calculations. The parameter `--dim 4 1 1` defines the supercell used to generate force constants.

When `--use_hiphive true` is selected, second- and third-order force constants are fitted from randomly displaced structures. The parameters `--n_structures`, `--rattle_std`, `--min_dist`, and `--cutoffs` control the number of training structures, displacement amplitude, minimum interatomic distance, and cutoff radii, respectively.

The q-point mesh and temperature range are given by `--mesh` and `--temps`. In this example, a one-dimensional mesh of `51 1 1` is used, and the thermal conductivity is calculated from 100 K to 1000 K with a step of 100 K. Finally, `--method lbte` means that the linearized Boltzmann transport equation is solved, while `--wigner false` indicates that the conventional phonon BTE rather than the Wigner transport equation is used.

S5. Sensitivity of the Monte Carlo baseline to boundary roughness

In the phonon Monte Carlo (MC) simulations, the periodically modulated cove-shaped boundary is explicitly resolved in the simulation geometry. Therefore, the root-mean-square roughness amplitude η in the Ziman–Soffer relation,

$$p_{\mathrm{spec}} = \exp\left[-4\eta^2 k^2 \cos^2\theta\right], \tag{1}$$

does not describe the cove modulation itself, but phenomenologically represents unresolved atomic-scale boundary non-specularity. Here, p_{spec} is the probability of specular boundary reflection, k is the magnitude of the incident phonon wave vector, and θ is the angle between the incident wave vector and the local boundary normal.

To examine the influence of η on the MC baseline, we performed calculations for a reference AGNR with $L = 10000$ nm and $W = 500$ nm, varying η from 0.06 to 0.14 nm while keeping all other simulation parameters unchanged. Three independent simulations were performed for each value.

As shown in Fig. S1, the MC thermal conductivity decreases monotonically with increasing η because of enhanced diffuse boundary scattering. The independently calculated MD thermal conductivity for the same AGNR is $1741\ \mathrm{W\,m^{-1}\,K^{-1}}$. Interpolation of the MC results gives an intersection with this value at approximately $\eta = 0.101$ nm, supporting the use of $\eta = 0.1$ nm in the main calculations.

The same $\eta = 0.1$ nm is subsequently applied to both AGNR and Cove-GNR simulations without further adjustment. Thus, their MC conductivity difference primarily reflects the explicitly resolved deterministic boundary geometries under a common boundary-scattering condition. We note that the quantitative MC baseline, and consequently the residual suppression defined from the MD–MC comparison, depends on the selected effective roughness parameter.

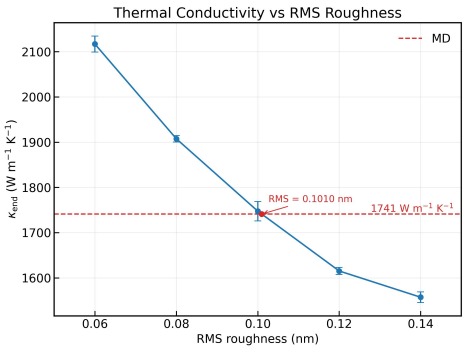


Figure S1: MC thermal conductivity of the reference AGNR ($L = 10000$ nm, $W = 500$ nm) as a function of the RMS boundary roughness amplitude η. Each point represents the mean of three independent simulations, with error bars indicating one standard deviation. The horizontal dashed line denotes the MD reference value of $1741\ \mathrm{W\,m^{-1}\,K^{-1}}$. Its intersection with the interpolated MC results occurs at approximately $\eta = 0.101$ nm.

S6. Effect of hydrogen edge passivation

The main text focuses on carbon-only, unpassivated AGNR and Cove-GNR structures in order to isolate the effect of the carbon-backbone geometry. In realistic graphene nanoribbons, however, edge atoms may be chemically passivated, for example by hydrogen termination. Since hydrogen passivation changes the local edge bonding environment, it may modify edge-related vibrational modes and the resulting phonon thermal transport.

To examine this effect, we performed additional calculations for representative bare and H-passivated AGNR and Cove-GNR structures. Figure S2(a) compares the phonon dispersion and density of states of bare and H-passivated AGNRs calculated using the NEP-89 potential after structural relaxation. Hydrogen passivation introduces additional high-frequency edge-related vibrational features associated with C–H bonding, while the low-frequency acoustic branches remain broadly similar. Since thermal transport in graphene nanoribbons is mainly governed by low-frequency phonons, the influence of hydrogen passivation on the overall thermal conductivity is expected to be moderate.

Figure S2(a) further compares the phonon dispersion of the H-passivated AGNR obtained from DFT and NEP-89 calculations, confirming that the potential captures the main vibrational features of the passivated structure. Figure S2(b) compares the room-temperature thermal conductivity κ_{xx} of bare and H-passivated AGNR and Cove-GNR systems. The thermal conductivity changes only slightly after hydrogen passivation for the representative systems considered here. In contrast, the Cove-GNR still shows a substantially lower thermal conductivity than the corresponding AGNR, indicating that the cove-shaped carbon-backbone geometry remains the dominant factor responsible for the thermal-transport suppression discussed in the main text.

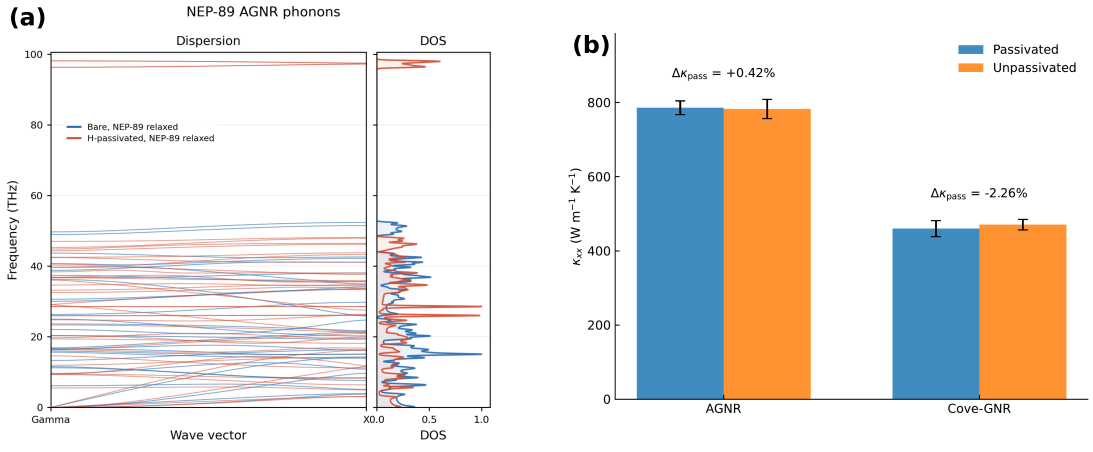


Figure S2: Effect of edge hydrogen passivation on the phonon transport properties of graphene nanoribbons. (a) Phonon dispersion and density of states of AGNRs with bare and H-passivated edges calculated using the NEP-89 potential after structural relaxation. (b) Comparison of the room-temperature thermal conductivity κ_{xx} of AGNR and Cove-GNR systems with bare and H-passivated edges. Error bars denote the statistical uncertainty.

S7. Statistical convergence of HNEMD thermal conductivity

The homogeneous nonequilibrium molecular dynamics (HNEMD) method evaluates the thermal conductivity of a periodically repeated system driven by a homogeneous external field. Therefore, the calculated thermal conductivity corresponds to the periodic-system response rather than a finite-length value obtained from explicit heat-source and heat-sink reservoirs. Unlike direct nonequilibrium MD, HNEMD does not require a length-convergence analysis along the transport direction. The main convergence requirement is the statistical convergence with respect to the production averaging time.

Figure S3 shows the accumulated thermal conductivity as a function of statistical averaging time. After the initial transient regime, the accumulated κ reaches a stable plateau, indicating that the production trajectories are sufficiently long for estimating the reported thermal conductivity values. In the main text, the error bars in Fig. 2 denote one standard deviation estimated from independent HNEMD simulations with different initial velocity seeds.

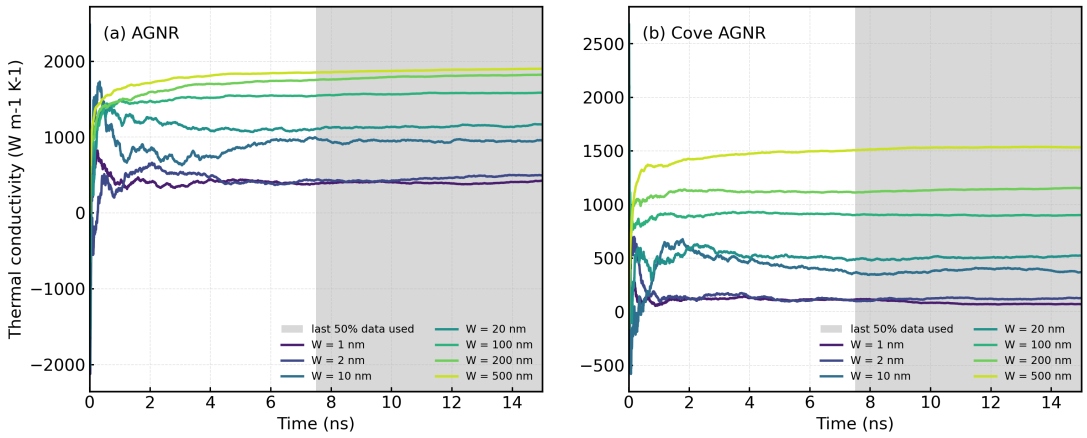


Figure S3: Statistical convergence of the HNEMD thermal conductivity as a function of averaging time. The accumulated thermal conductivity reaches a stable plateau during the production trajectory, confirming that the reported values are statistically converged.

S8. Mode-resolved vibrational-weight and edge-enrichment analysis

To characterize the spatial distribution of the representative Cove-GNR phonon modes, we calculated the normalized atomic vibrational weight

$$w_{i,\lambda} = \frac{\sum_{\alpha=x,y,z} |e^{\lambda}_{i\alpha}|^2}{\sum_j \sum_{\alpha=x,y,z} |e^{\lambda}_{j\alpha}|^2}, \tag{2}$$

where $e^{\lambda}_{i\alpha}$ is the eigenvector component of atom i in mode λ. The weights satisfy $\sum_i w_{i,\lambda} = 1$. In Fig. S8(a)–S8(c), we plot $Nw_{i,\lambda}$ so that a value of unity corresponds to a spatially uniform mode in a unit cell containing N atoms.

The edge-weight fraction and edge-enrichment factor are defined as

$$F_{\mathrm{edge},\lambda} = \sum_{i \in \mathrm{edge}} w_{i,\lambda}, \qquad E_{\mathrm{edge},\lambda} = \frac{F_{\mathrm{edge},\lambda}}{N_{\mathrm{edge}}/N}, \tag{3}$$

where the edge region contains the outermost two atomic layers adjacent to the cove boundaries. A uniform spatial distribution gives $E_{\mathrm{edge},\lambda} \approx 1$, while larger values indicate enhanced vibrational weight near the edges.

For the three modes shown in Fig. S4, the edge-weight fractions are $F_{\mathrm{edge}} = 0.165$, 0.233, and 0.837, corresponding to $E_{\mathrm{edge}} = 1.10$, 1.56, and 5.58, respectively. The third mode therefore has about 84% of its vibrational weight in the edge region, although this region contains only 15% of the atoms, indicating pronounced confinement near the cove edges.

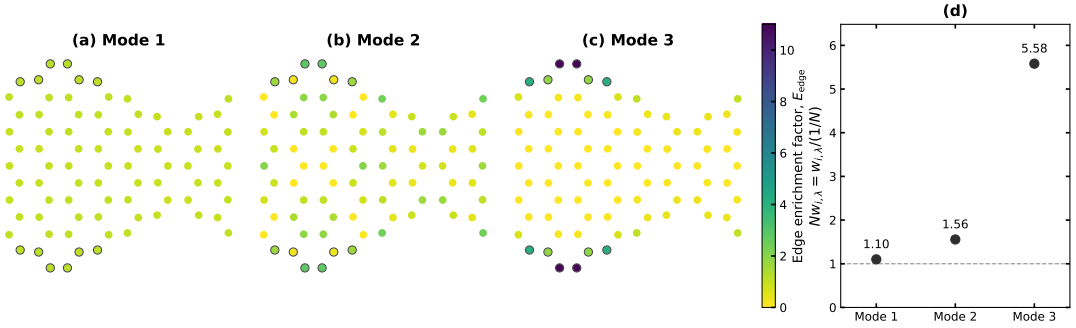


Figure S4: Mode-resolved vibrational-weight and edge-enrichment analysis of representative Cove-GNR phonon modes. (a–c) Normalized atomic vibrational-weight distributions $Nw_{i,\lambda}$ for the three representative modes shown in Fig. 3. (d) Edge-enrichment factor $E_{\mathrm{edge}} = F_{\mathrm{edge}}/(N_{\mathrm{edge}}/N)$. The dashed line denotes the uniform-distribution reference $E_{\mathrm{edge}} = 1$.

S9. Low-participation-mode statistics

To quantify the participation-ratio difference between AGNR and Cove-GNR, we calculated the fraction of phonon modes with strongly reduced modal participation. The low-participation-mode fraction is defined as

$$f_{\text{low-PR}} = \frac{N(\text{PR} < 0.3)}{N_{\text{total}}}, \tag{4}$$

where $N(\text{PR} < 0.3)$ is the number of phonon modes with $\text{PR} < 0.3$, and N_{total} is the total number of modes included in the analysis. The same q-point sampling and frequency range were used for the AGNR and Cove-GNR comparison.

For the 1-nm-wide nanoribbons, the AGNR contains 133 low-PR modes out of 2754 total modes, corresponding to $f_{\text{low-PR}} = 4.83\%$. In contrast, the Cove-GNR contains 2174 low-PR modes out of 12240 total modes, corresponding to $f_{\text{low-PR}} = 17.76\%$. The Cove-GNR therefore has a substantially larger population of low-participation modes than the pristine AGNR.

This quantitative statistic is consistent with the participation-ratio distributions and representative eigenvectors shown in Fig. 3 of the main text. It further shows that cove-edge modulation increases the fraction of phonon modes with reduced spatial participation.

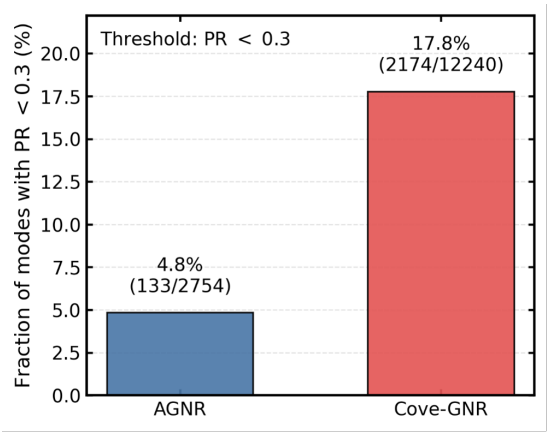


Figure S5: Low-participation-mode fraction in 1-nm-wide AGNR and Cove-GNR. The Cove-GNR shows a larger low-PR fraction, 17.76%, than the AGNR, 4.83%, indicating that cove-edge modulation increases the population of spatially confined vibrational modes.

S10. Effective phonon mean free path

To characterize the phonon transport length scale in AGNRs and Cove-GNRs, we evaluated the effective phonon mean free path based on the MD size-dependent transport analysis. For each ribbon width, the length-dependent results were used to obtain an averaged transport mean free path for the corresponding ribbon geometry.

We compared the dimension-dependent mean free path under two fixed conditions. At fixed $W = 20$ nm, the mean free path of the Cove-GNR remains lower than that of the AGNR over the examined length range. At fixed $L = 1000$ nm, the mean free path increases with ribbon width for both structures, while the Cove-GNR consistently shows a shorter mean free path than the AGNR.

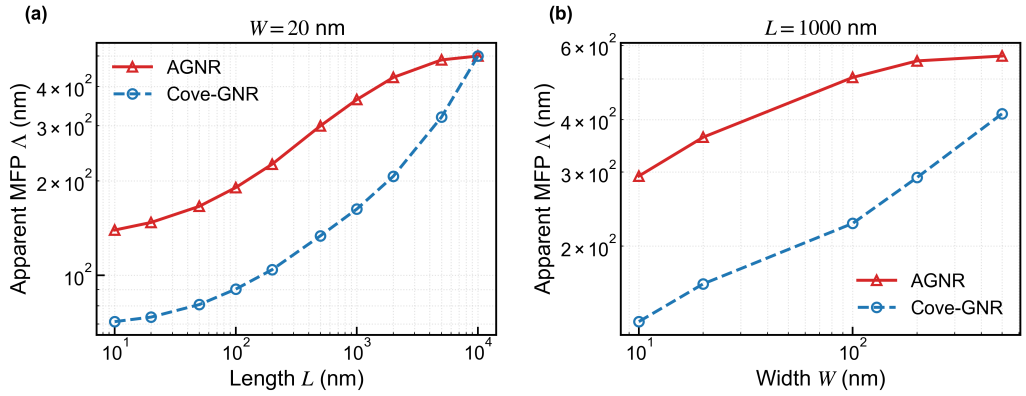


Figure S6: Effective phonon mean free path of AGNR and Cove-GNR obtained from the MD-based size-dependent transport analysis. (a) Mean free path as a function of ribbon length at fixed width $W = 20$ nm. (b) Mean free path as a function of ribbon width at fixed length $L = 1000$ nm.